MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 1)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The distance of the point $P(3, 4)$ from the origin is: [1]

(A) 3 **(B)** 4 **(C)** 5 **(D)** 7

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 5

Given: Point $P(3, 4)$ and origin $O(0, 0)$.

Formula Used: Distance from origin $= \sqrt{x^2 + y^2}$

Substituting $x = 3$ and $y = 4$:

$$OP = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = \boxed{5 \text{ units}}$$

Q2. The distance of the point $P(-6, 8)$ from the origin is: [1]

(A) 8 **(B)** $2\sqrt{7}$ **(C)** 10 **(D)** 6

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 10

Given: Point $P(-6, 8)$ and origin $O(0, 0)$.

Formula Used: Distance from origin $= \sqrt{x^2 + y^2}$

Substituting $x = -6$ and $y = 8$:

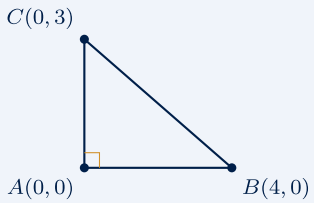
$$OP = \sqrt{(-6)^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = \boxed{10 \text{ units}}$$

Q3. ABC is a triangle. B lies on the x-axis at a distance of 4 units from the y-axis at its right. C is on the y-axis and is 3 units away from the origin. If the coordinates of A are $(0, 0)$, the perimeter of $\triangle ABC$ is: [1]

(A) 7 units **(B)** 12 units **(C)** 6 units **(D)** 15 units

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 12 units



Given: Vertex $A(0, 0)$, $B(4, 0)$ (to the right of y-axis on x-axis), and $C(0, 3)$ (or $(0, -3)$ on y-axis). We compute all three sides and sum them.

We apply the distance formula for perimeter:

Formula Used: Perimeter $= AB + BC + AC$, $\quad d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Computing the three side lengths:

$$AB = 4 \text{ units}, \qquad AC = 3 \text{ units}$$

$$BC = \sqrt{(0 - 4)^2 + (\pm 3 - 0)^2} = \sqrt{16 + 9} = \sqrt{25} = 5 \text{ units}$$

$$\text{Perimeter} = 4 + 5 + 3 = \boxed{\mathbf{12} \text{ units}}$$

Q4. The distance of the point $P(2, -2)$ from the origin is: [1]

(A) 2 **(B)** -2 **(C)** 0 **(D)** $2\sqrt{2}$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D) $2\sqrt{2}$

Given: Point $P(2, -2)$ and origin $O(0, 0)$.

Formula Used: Distance from origin $= \sqrt{x^2 + y^2}$

Substituting $x = 2$ and $y = -2$:

$$OP = \sqrt{2^2 + (-2)^2} = \sqrt{4 + 4} = \sqrt{8} = \sqrt{4 \times 2} = \boxed{\mathbf{2\sqrt{2}} \text{ units}}$$

Q5. The distance of the point $(-1, 7)$ from the x-axis is: [1]

(A) -1 **(B)** 7 **(C)** 6 **(D)** $\sqrt{50}$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 7

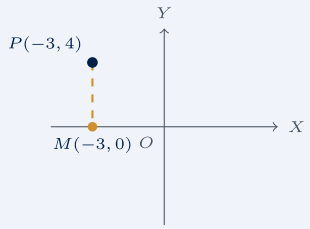
Given: Point $P(-1, 7)$. Foot of perpendicular on the x-axis is $M(-1, 0)$.

Formula Used: Perpendicular distance of (x, y) from x-axis $= |y|$

$$PM = \sqrt{(-1 - (-1))^2 + (0 - 7)^2} = \sqrt{0 + 49} = |7| = \boxed{\mathbf{7} \text{ units}}$$

Q6. Find the distance of the point $P(-3, 4)$ from the x-axis. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Coordinate of Perpendicular Foot [1 Mark]
Given: Point $P(-3, 4)$. The point directly below it on the x-axis is $M(-3, 0)$.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 2: Applying the Distance Formula [2 Marks]

$$PM = \sqrt{(-3 - (-3))^2 + (0 - 4)^2} = \sqrt{0^2 + (-4)^2} = \sqrt{16} = \boxed{4 \text{ units}}$$

Q7. Find the distance between the points $A(2, 3)$ and $B(4, 1)$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Formula Declaration [1 Mark]
Given: $A(x_1, y_1) = (2, 3)$ and $B(x_2, y_2) = (4, 1)$.

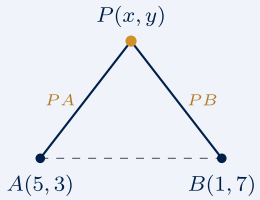
Formula Used: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 2: Substitution and Simplification [2 Marks]

$$AB = \sqrt{(4 - 2)^2 + (1 - 3)^2}$$
$$= \sqrt{2^2 + (-2)^2} = \sqrt{4 + 4} = \sqrt{8} = \sqrt{4 \times 2} = \boxed{\mathbf{2\sqrt{2}} \text{ units}}$$

Q8. Find a relation between x and y such that the point $P(x, y)$ is equidistant from the points $A(5, 3)$ and $B(1, 7)$. [5]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Equating Distances ($PA = PB$) [1.5 Marks]
Given: $P(x, y)$ is equidistant from $A(5, 3)$ and $B(1, 7)$, so $PA = PB \implies PA^2 = PB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Algebraic Expansion [2 Marks]

$$(x-5)^2 + (y-3)^2 = (x-1)^2 + (y-7)^2$$

$$(x^2 - 10x + 25) + (y^2 - 6y + 9) = (x^2 - 2x + 1) + (y^2 - 14y + 49)$$

$$x^2 + y^2 - 10x - 6y + 34 = x^2 + y^2 - 2x - 14y + 50$$

Step 3: Simplification to Linear Relation [1.5 Marks]
Cancelling $x^2 + y^2$ from both sides:

$$-10x - 6y + 34 = -2x - 14y + 50$$

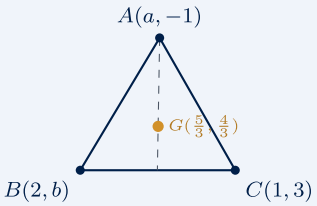
$$-10x + 2x - 6y + 14y = 50 - 34 \implies -8x + 8y = 16$$

Dividing throughout by -8:

$$\boxed{\mathbf{x - y = -2} \quad (\text{or } \mathbf{y - x = 2})}$$

Q9. If $G\left(\frac{5}{3}, \frac{4}{3}\right)$ is the centroid of $\triangle ABC$ whose vertices are $A(a, -1)$, $B(2, b)$ and $C(1, 3)$, find the values of a and b. [4]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]



Step 1: Centroid Formula Declaration [1 Mark]
Given: Vertices $A(a, -1)$, $B(2, b)$, $C(1, 3)$ and centroid $G\left(\frac{5}{3}, \frac{4}{3}\right)$.

We apply the centroid formula:

Formula Used: $G(x, y) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$

Step 2: Substitution and Solving for a and b [3 Marks]

$$\left(\frac{a+2+1}{3}, \frac{-1+b+3}{3}\right) = \left(\frac{5}{3}, \frac{4}{3}\right) \implies \left(\frac{a+3}{3}, \frac{b+2}{3}\right) = \left(\frac{5}{3}, \frac{4}{3}\right)$$

Equating the x-coordinates:

$$\frac{a+3}{3} = \frac{5}{3} \implies a + 3 = 5 \implies a = 5 - 3 = \mathbf{2}$$

Equating the y-coordinates:

$$\frac{b+2}{3} = \frac{4}{3} \implies b + 2 = 4 \implies b = 4 - 2 = \mathbf{2}$$

$$\boxed{\mathbf{a = 2} \quad \text{and} \quad \mathbf{b = 2}}$$

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